Internet of Things

Dr. Ndeh Ningo

Internet of Things

Phases in the Evolution of the Internet

Internet of Things

Phases in the Evolution of the Internet

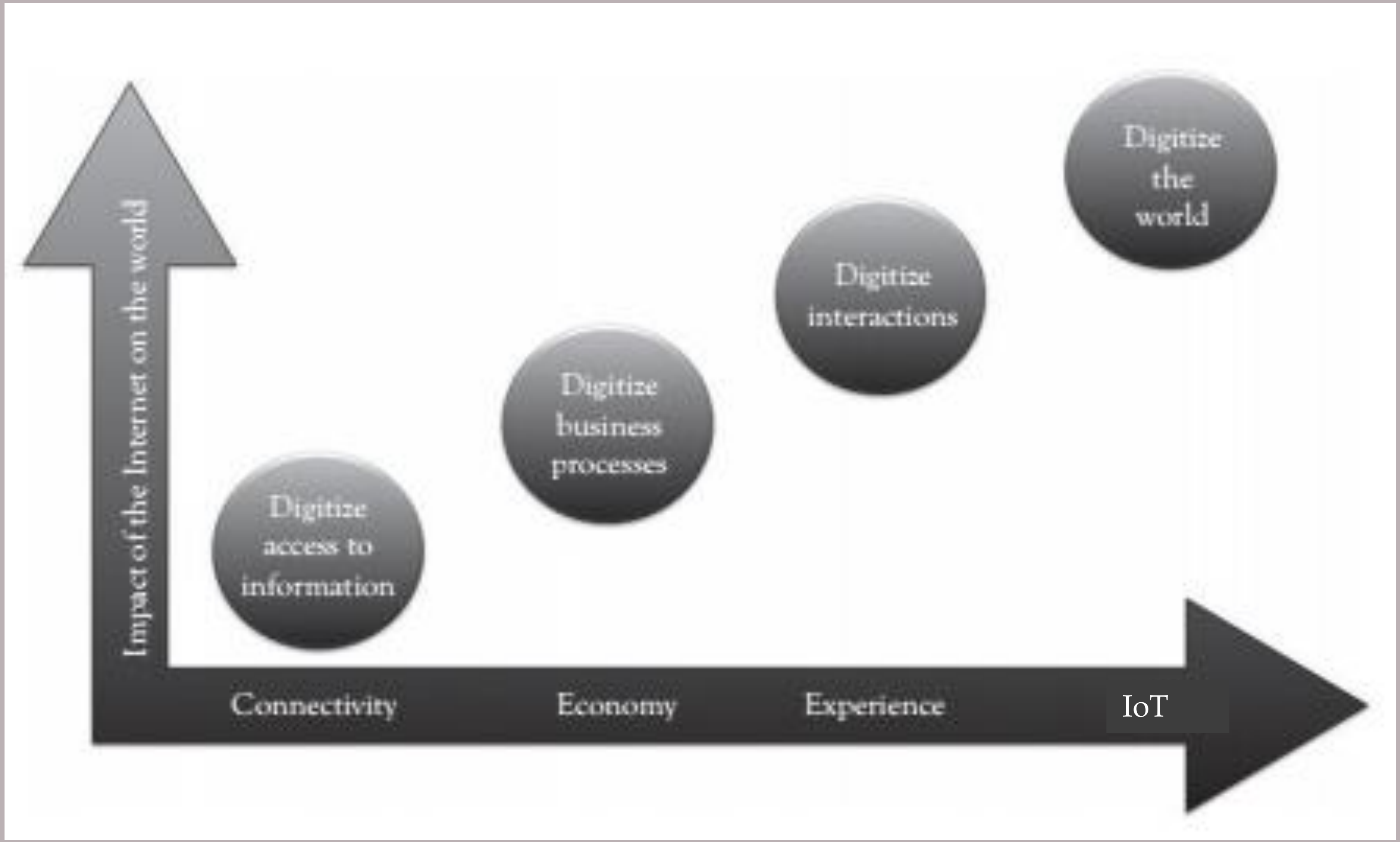
To fully understand the Internet of Things and its overwhelming potential impact on all aspects of society, it is necessary to begin by tracing the evolution of the Internet to establish the unavoidable link to the Internet of Thing.

Internet of Things

Phases in the Evolution of the Internet

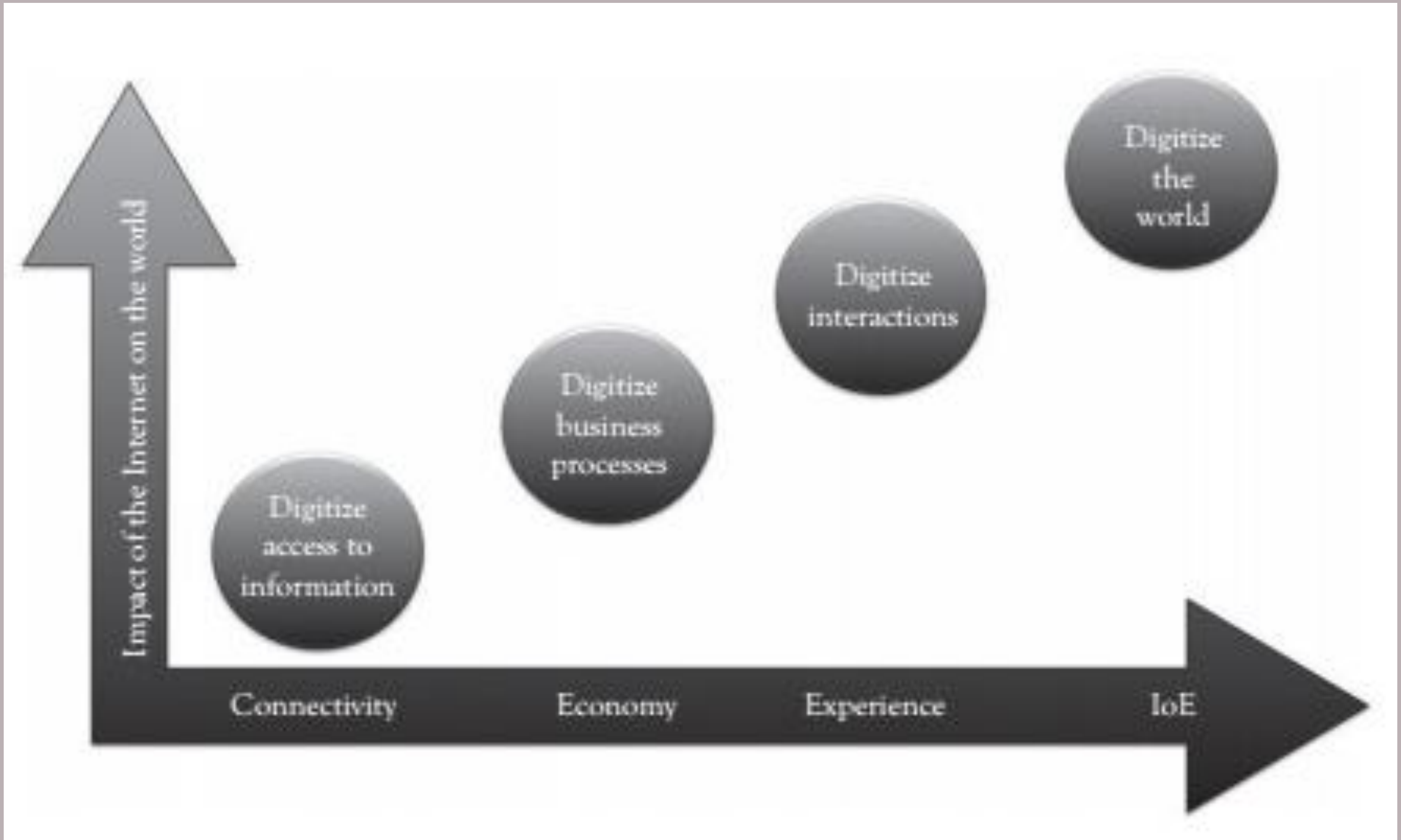
Looking at the evolution of the Internet, we can identify four very distinct phases or waves.

Internet of Things



Four phases in the evolution of the Internet

Internet of Things



Four phases in the evolution of the Internet

Each of these four phases builds on the previous one, and the technological and social advances of the past few decades have led to IoT.

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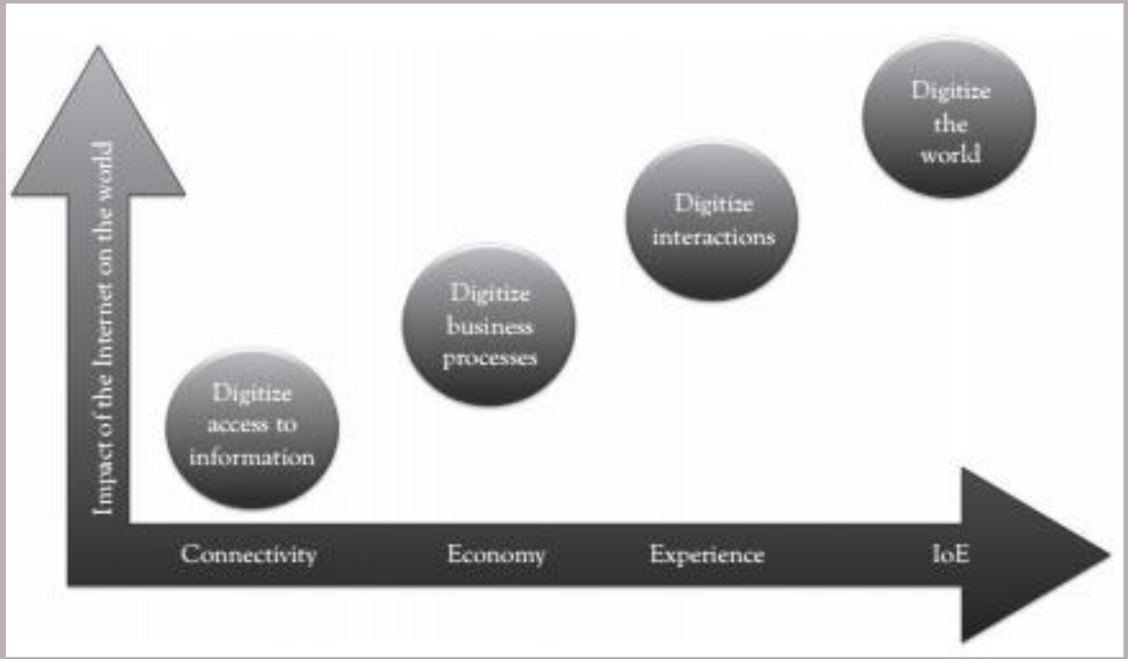
Phases in the Evolution of the Internet

Each phase has a significantly more profound effect on business and society.



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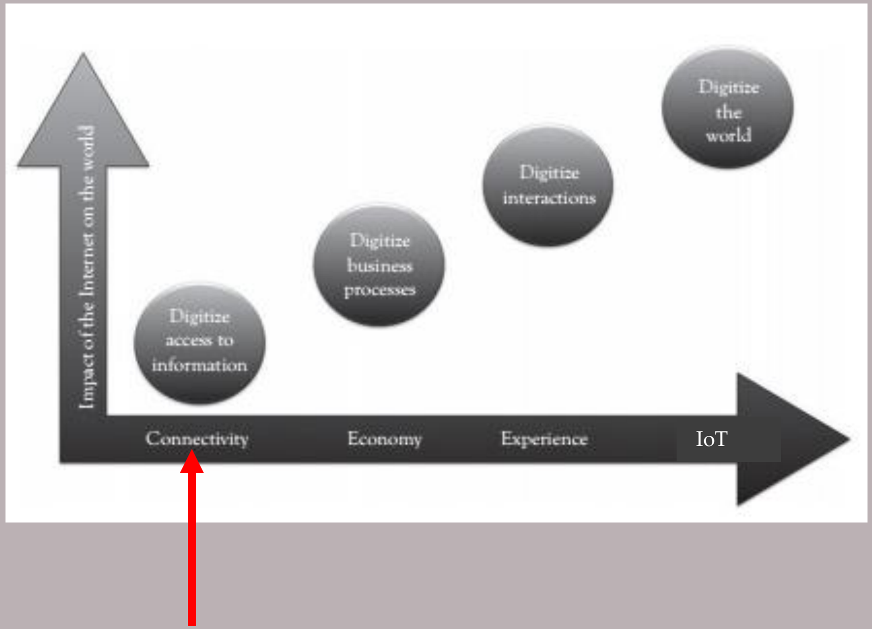
Phases in the Evolution of the Internet

1. In the first phase, which began in the mid 1990's, the goal is to achieve *networked connectivity*, and consequently, *digitizing the access to information*.

   A couple of decades ago, just getting connected to the online world was miraculous enough. The World Wide Web (WWW) flourished, and the distance and time to access information dramatically shrank. People looked for information online, sent e-mils, and searched the Web.



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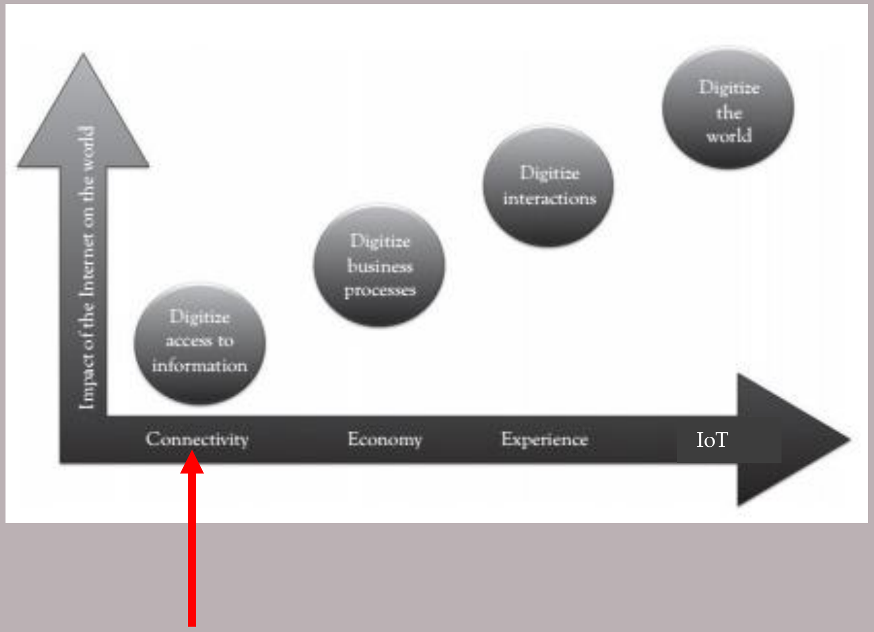
Phases in the Evolution of the Internet

1. In the first phase, the goal was to achieve *networked connectivity*, and consequently, *digitizing the access to information*.



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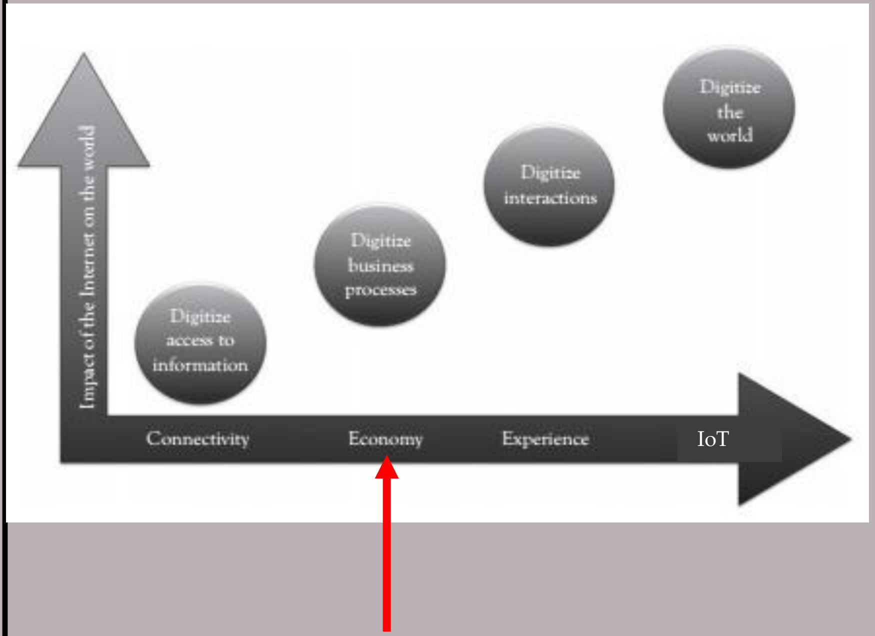
Phases in the Evolution of the Internet

2. The second phase is characterized as the *networked economy*, in which business processes were digitized. People bought products online. During the networked economy phase, e-commerce was developed and the supply chain was digitally connected providing additional digital consolidation.



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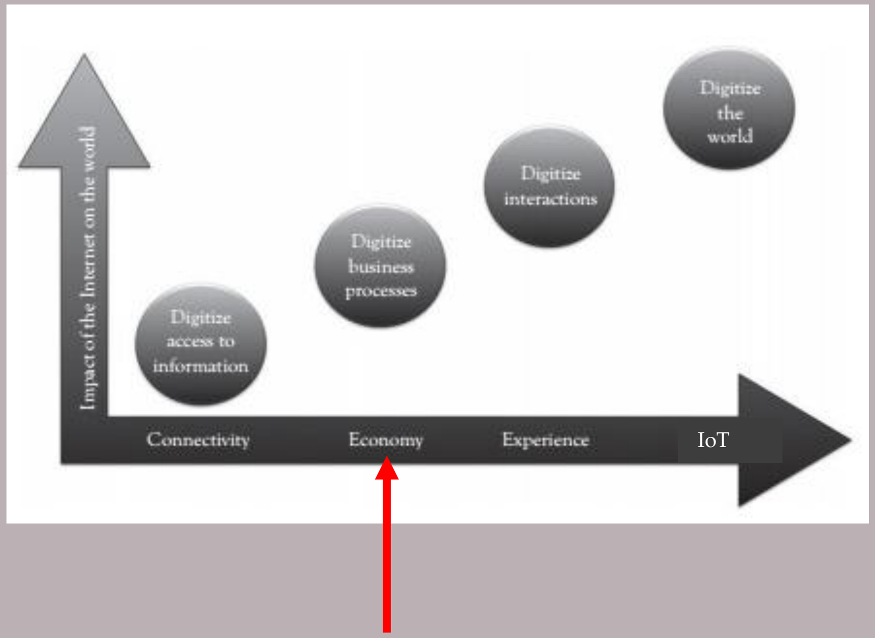
Phases in the Evolution of the Internet

2. With the *Networked Economy*, e-commerce and digitally connected supply chains caused one of the major disruptions of the past 120 years. Vendors and suppliers became closely interlinked with producers, and online shopping experienced incredible growth.

The victims of this shift were traditional brick-and-mortar retailers. The economy itself became more digitally intertwined as suppliers, vendors, and consumers all became more directly connected.



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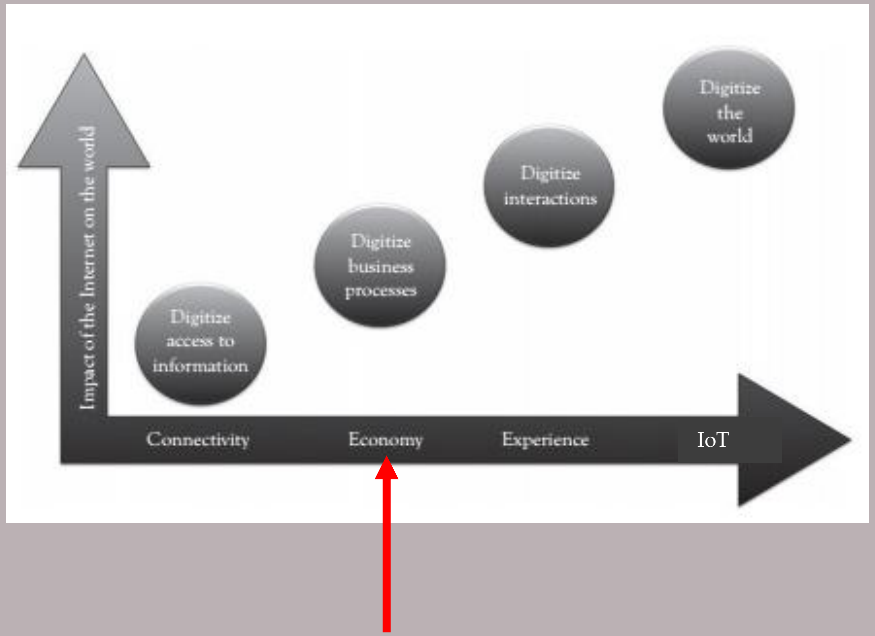
Phases in the Evolution of the Internet

2. This phase of the *Networked Economy* enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.



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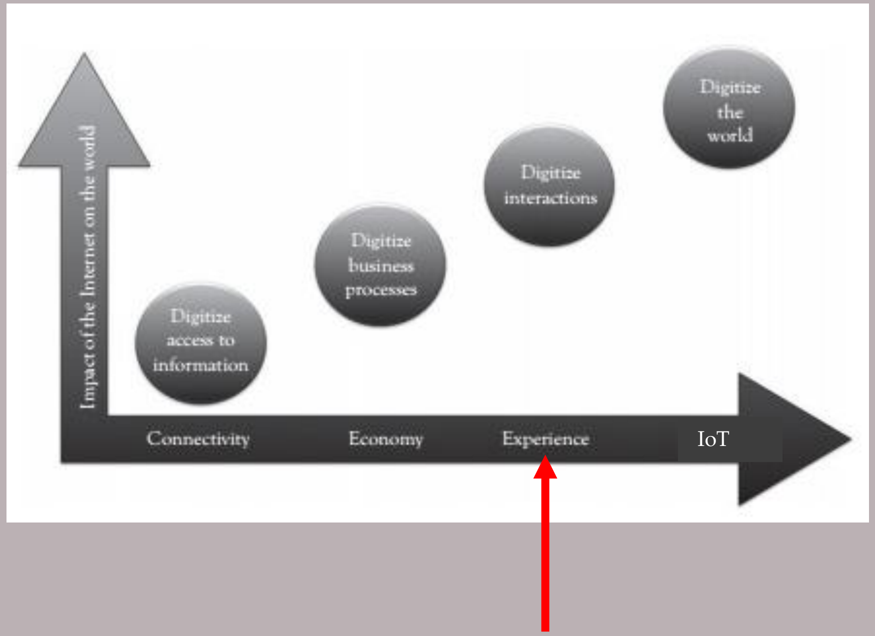
Phases in the Evolution of the Internet

3. In the third phase, *networked emerging experiences*, interactions were digitized. These digitized experiences belonged to both the business and personal realms. This phase was dominated by digitally enabled social media and collaboration. Furthermore, widespread access to digital mobility compounded the effects of this phase where digital video for personal interactions was pervasive. People connected digitally with each other.



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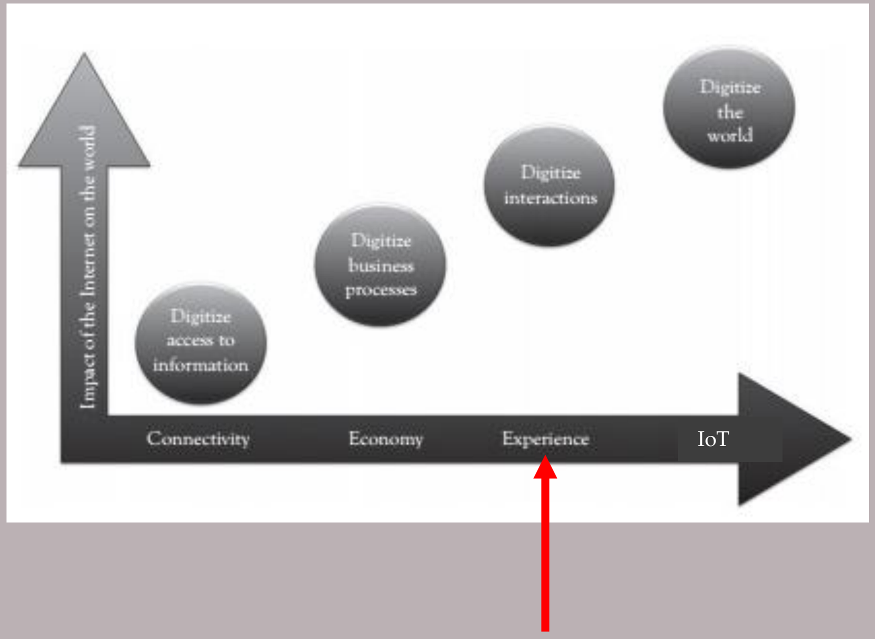
Phases in the Evolution of the Internet

3. The third phase, *Immersive Experiences*, was characterized by the emergence of social media, collaboration, and widespread mobility on a variety of devices. Connectivity was now pervasive, using multiple platforms from mobile phones to tablets to laptops and desktop computers.

This pervasive connectivity in turn enabled communication and collaboration as well as social media across multiple channels, via email, texting, voice, and video. In essence, person-to-person interactions became digitized.



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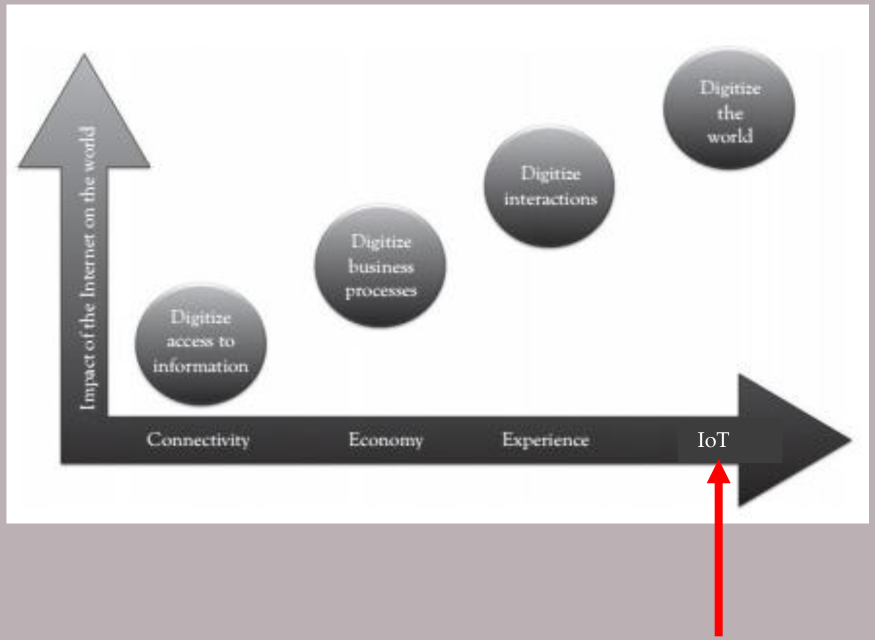
Phases in the Evolution of the Internet

4. The fourth phase, which we are entering, is referred to as the *Internet of Things*.

   In this phase the whole world is digitized! We are entering a phase that unlocks an unprecedented value for businesses, governments, and society in general.



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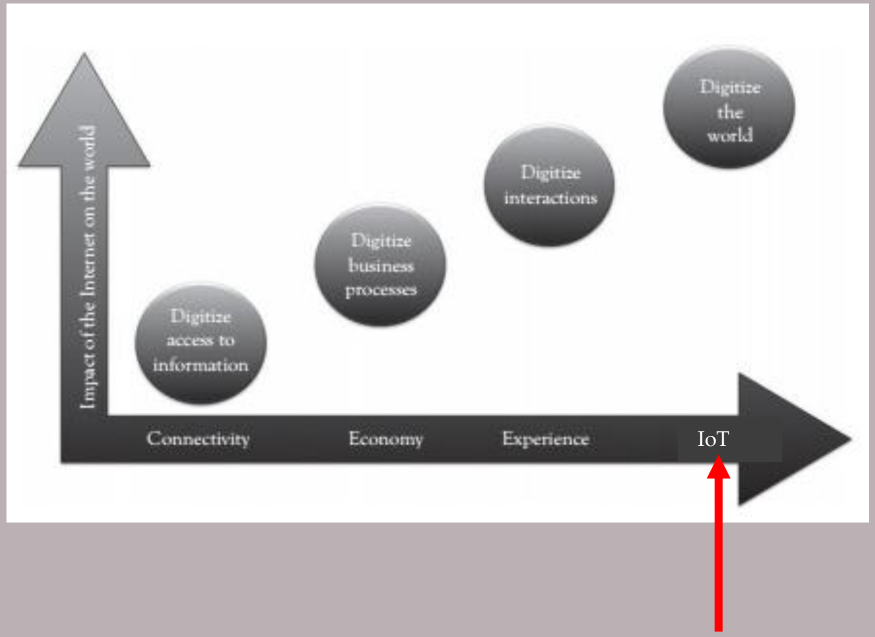
Phases in the Evolution of the Internet

4. The latest phase is the *Internet of Things*. Despite all the talk and media coverage of IoT, in many ways we are just at the beginning of this phase. When you think about the fact that 99% of “things” are still unconnected, you can better understand what this evolutionary phase is all about.



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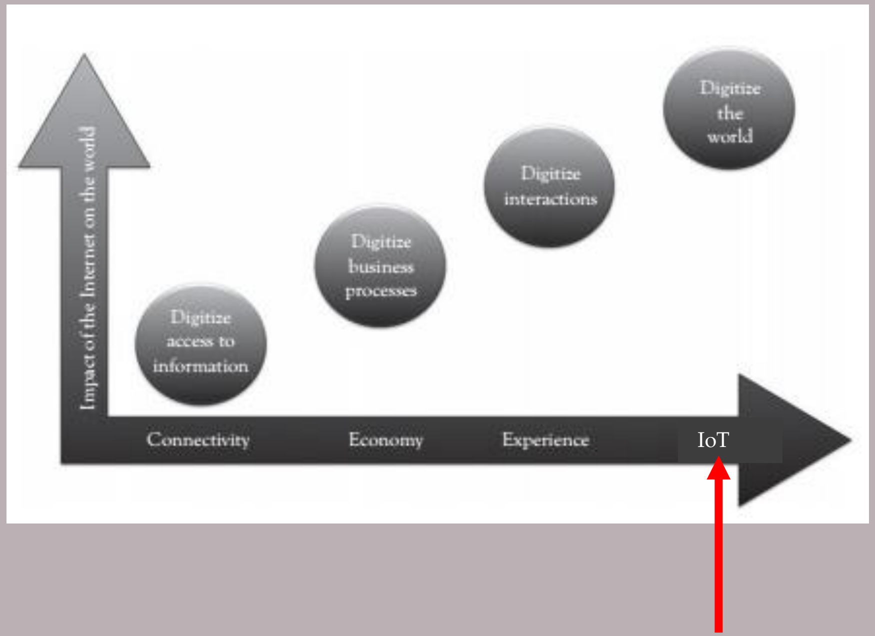
Phases in the Evolution of the Internet

4. *Internet of Things :* Machines and objects in this phase connect with other machines and objects, along with humans. Business and society have already started down this path and are experiencing huge increases in data and knowledge.



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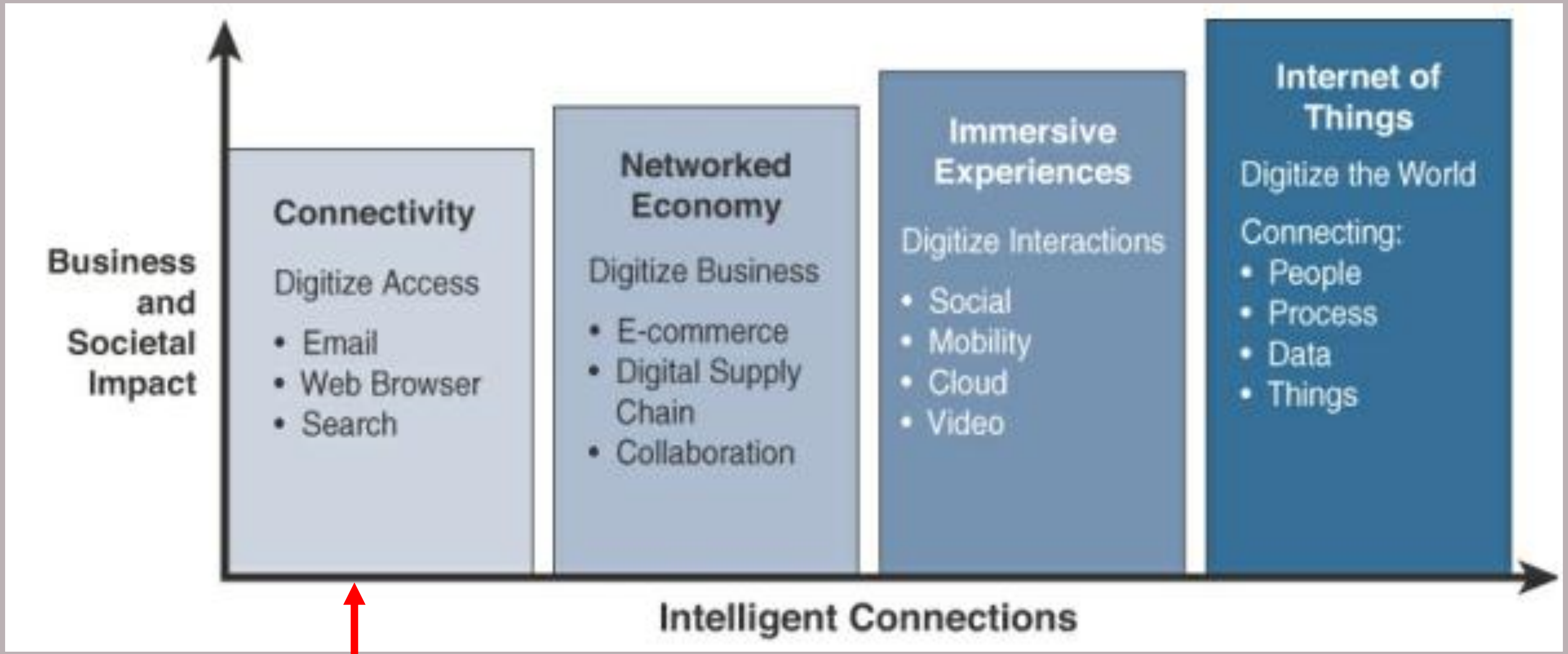
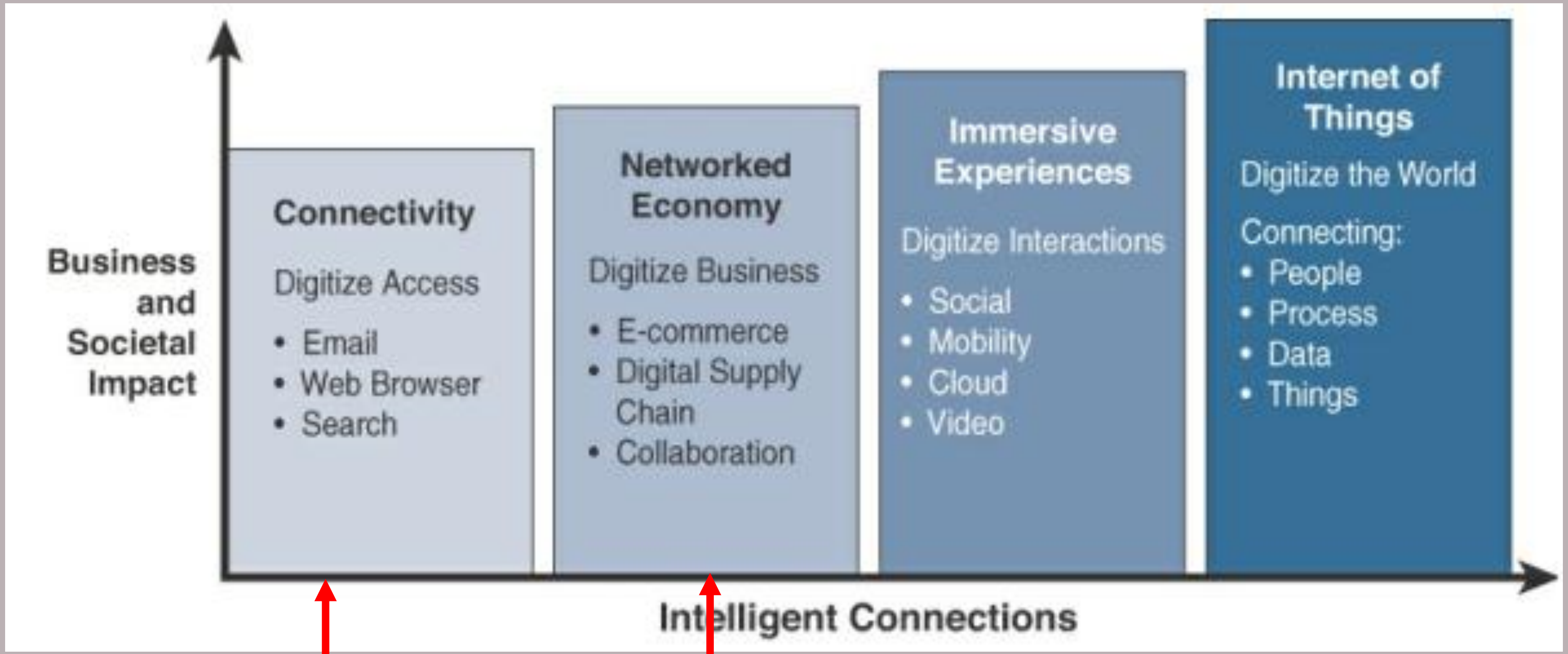


Evolutionary Phases of the Internet

Connectivity (Digitize access): This phase connected people to email, web services, and search so that information is easily accessed.

Internet of Things

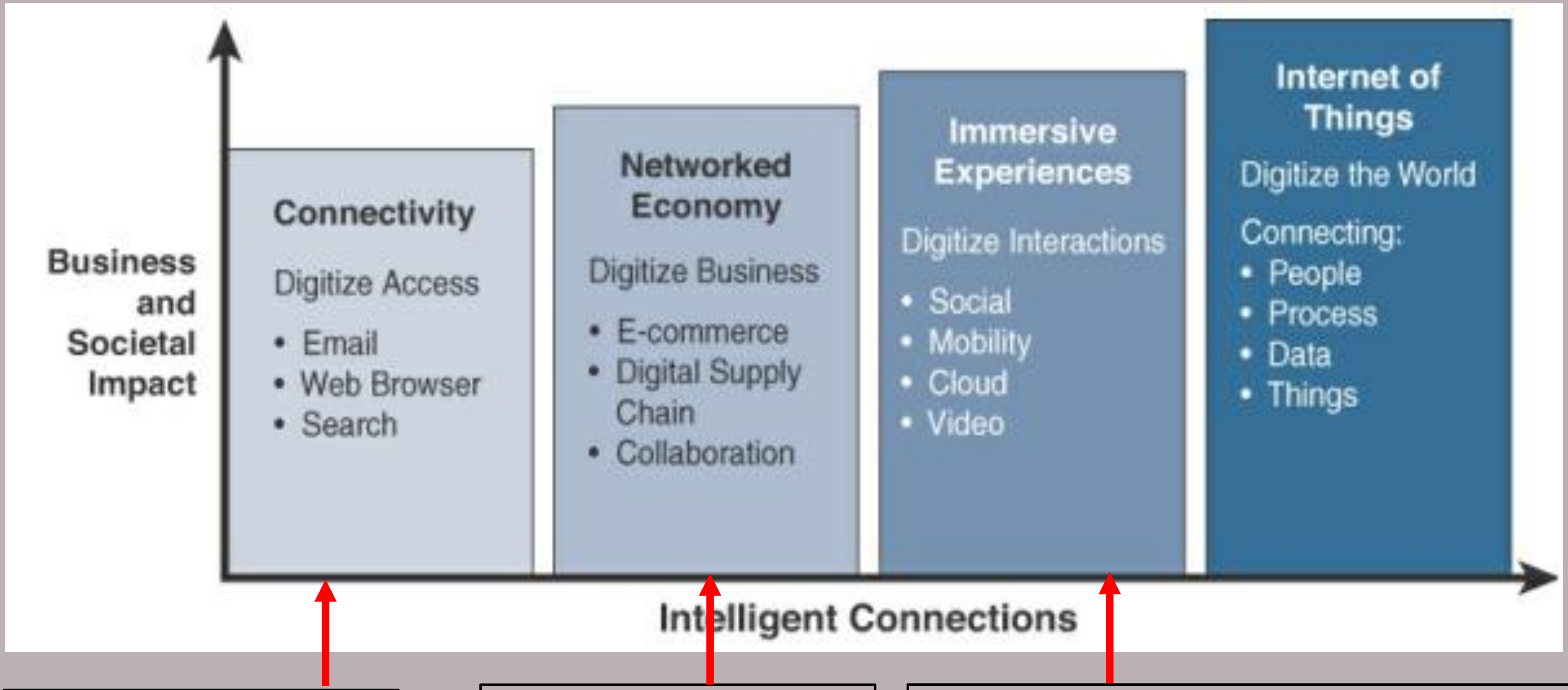


Evolutionary Phases of the Internet

Connectivity (Digitize access)

Networked Economy (Digitize business): e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.

Internet of Things



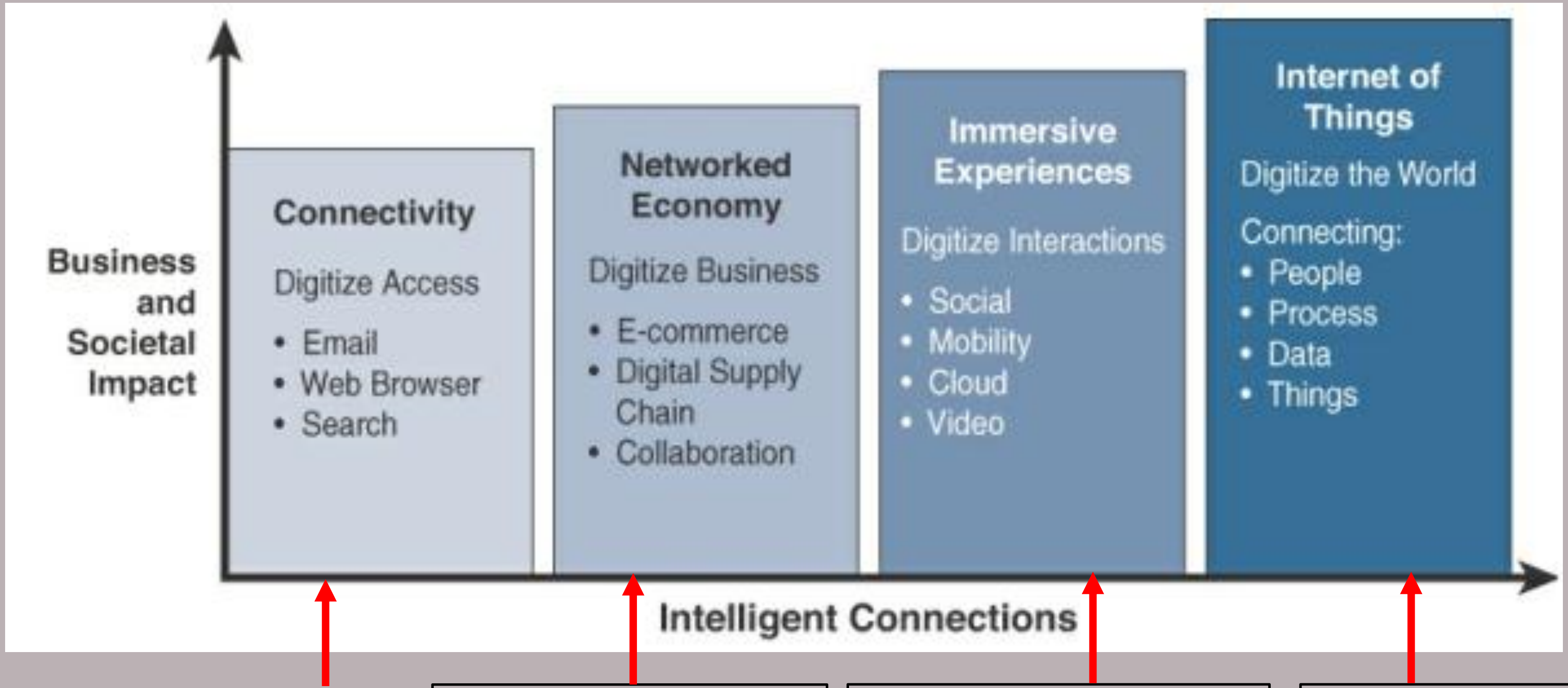
Evolutionary Phases of the Internet

Connectivity (Digitize access)

Networked Economy (Digitize business)

Immersive Experiences (Digitize interactions): Internet experience extended to encompass widespread video and social media while always being connected through mobility. More and more applications are into the cloud.

Internet of Things



Evolutionary Phases of the Internet

Connectivity (Digitize access)

Networked Economy (Digitize business)

Immersive Experiences (Digitize interactions)

Internet of Things (Digitize the world): connectivity is extended to objects (things) and machines in the world around us to enable new services and experiences. It is connecting the unconnected.

Internet of Things

- The growth of the Internet is occurring in accelerated waves. The acceleration is often explained as following Metcalf's Law, or the *network effect*, which explains that the value of a network grows exponentially with the number of endpoints being connected (or proportionately to the square of the number of users).
- In other words, the more *things* and people connect, the larger the value of the network.

Internet of Things

Internet of Computers

Versus

Internet of Things

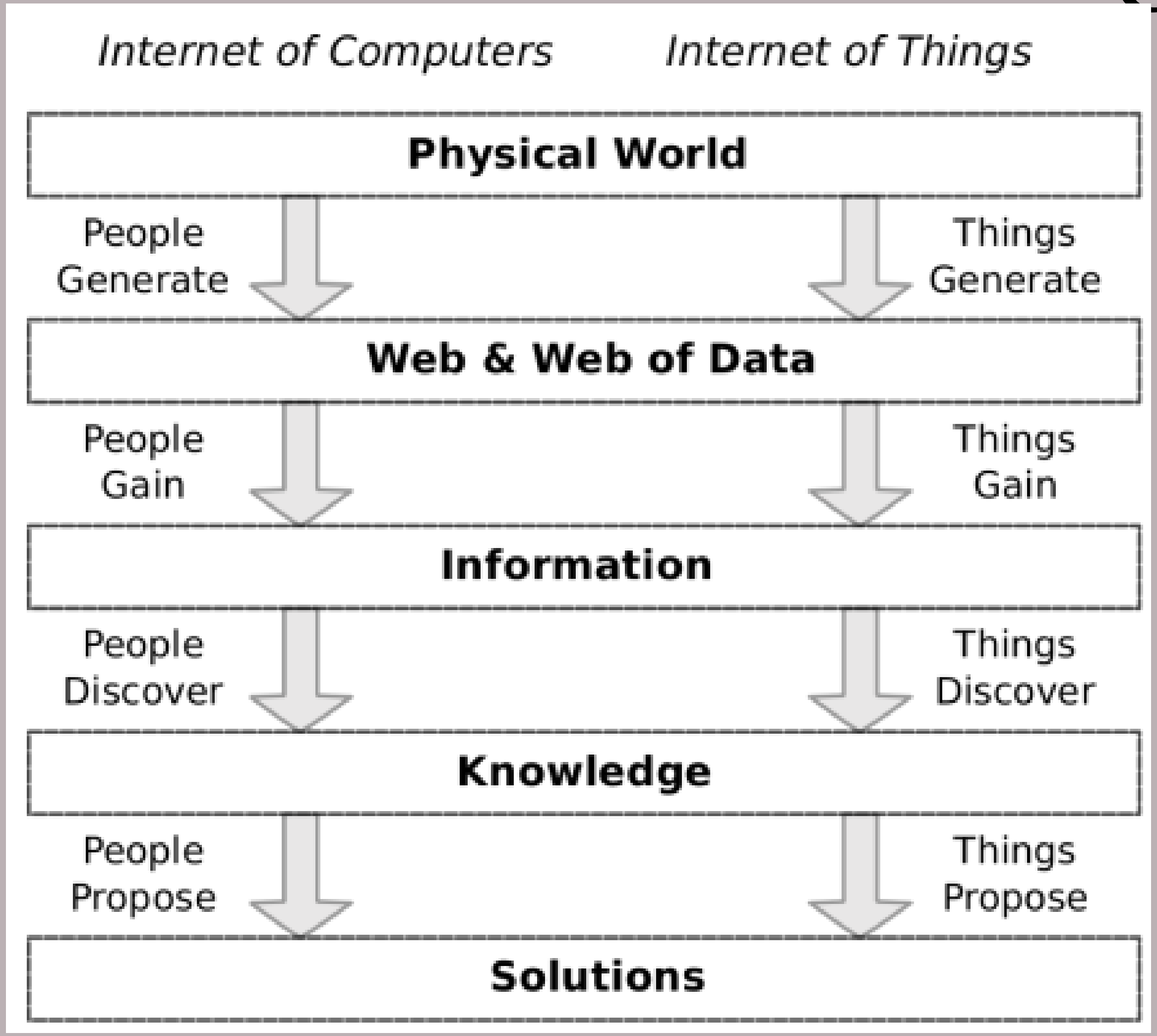
Internet of Things

Introduction

- The Internet is a global system of networks interconnecting computers using the standard Internet protocol suite. It is a network of networks and each network connects various numbers of computers. Hence, the traditional Internet has a focus on ***computers*** and can be called the Internet of Computers.
- In contrast, evolving from the Internet of Computers, the Internet of Things (IoT) emphasizes ***things*** rather than computers. It aims to connect everyday objects, such as coats, shoes, watches, ovens, washing machines, bikes, cars, even humans, plants, animals, and changing environments, to the Internet to enable communication/interactions between these objects.

Internet of Things

Introduction



Internet of Computers
versus
Internet of Things

Ubiquitous Network

Anytime, Anywhere, by Anyone and Anything

Internet of Things

Introduction

- In the context of the Internet, addressable and interconnected things, instead of humans, act as the main data producers, as well as the main data consumers.
- Computers will be able to learn and gain information and knowledge to solve real world problems directly with the data fed from things.
- As an ultimate goal, computers enabled by the Internet of Things technologies will be able to sense and react to the real world for humans.

Internet of Things

Introduction

- In this perspective, the conventional concept of the Internet as an infrastructure network reaching out to end-users' terminals will fade, leaving space to a notion of interconnected **"smart" objects** forming **pervasive computing** environments.
- The Internet infrastructure will not disappear. On the contrary, it will retain its vital role as global backbone for worldwide information sharing and diffusion, interconnecting physical objects with computing/communication capabilities across a wide range of services and technologies.

Internet of Things

Introduction

- This innovation will be enabled by the **embedding of electronics** into everyday **physical objects**, making them **"smart"** and letting them seamlessly **integrate within the global resulting cyber-physical infrastructure**.
- The Internet of Things (IoT) requires the development of advanced techniques able to embed computing, communication, sensing and identification capabilities into everyday objects.

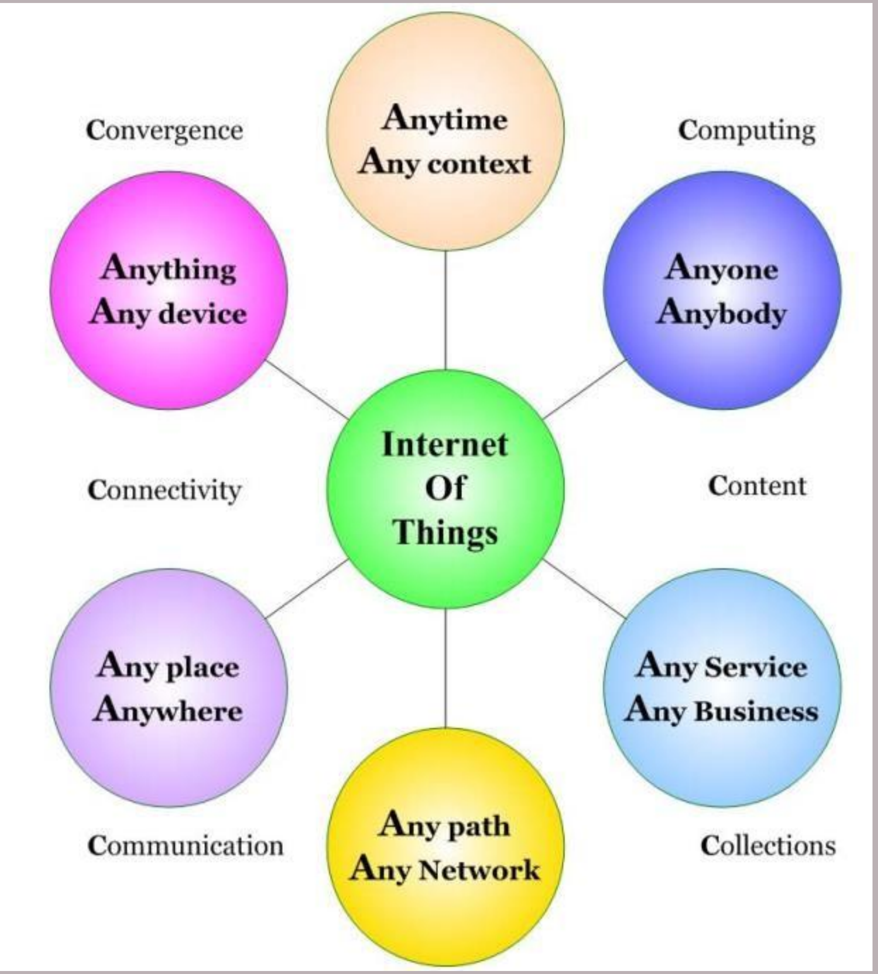
Internet of Things

"Connected Any-time, Any-place, with Any-thing and Any-one

Ideally using Any-path/network and Any-service"

Internet of Things

The basic idea of IoT can be conceived as a representation of various A's and C's, as shown in the figure in which the A's reflect the concept of ubiquity or globalization (i.e. any device, anywhere, anytime, any network etc.) and the C's mirror the main characteristics of IoT (i.e. connectivity, computing, convergence, etc.).



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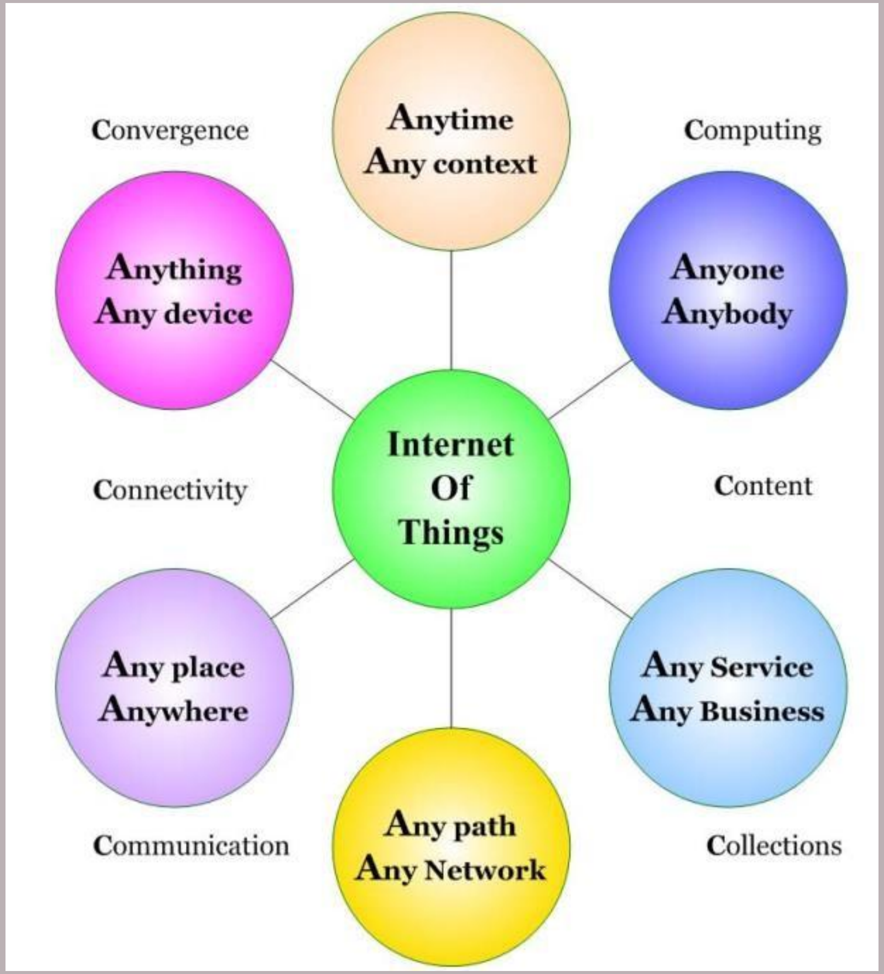
IoT developments during recent years have been characterized by attributes that can be "labelled" the 6**A**'s:
Anything (any device), to be transferred from/to Anyone (anybody), located Any place (anywhere), at Any time (any context), using the most appropriate physical path from Any path (any network) available between the sender and the recipient based on performance and/or economic considerations, to provide Any service (any business).



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The IoT paradigm is evolving and entire IoT ecosystems are now built upon innervation elements known as the 6**C's**:

- **C**ollect (heterogeneity of devices of various complexities and intelligence, that enhance the real-time collection of data generated from the connections of devices and information),
- **C**onnect (ubiquitous distributed connections of heterogeneous devices and information, where the connections are the foundational component of the IoT),
- **C**ache (stored information in the distributed IoT computing/processing environment),
- **C**ompute (advanced processing and computation of data and information),
- **C**ognize (information analytics, insights, extractions, real-time AI processing and
- **C**reate (the creation of new interactions, services, experiences, business models and solutions).

Internet of Things

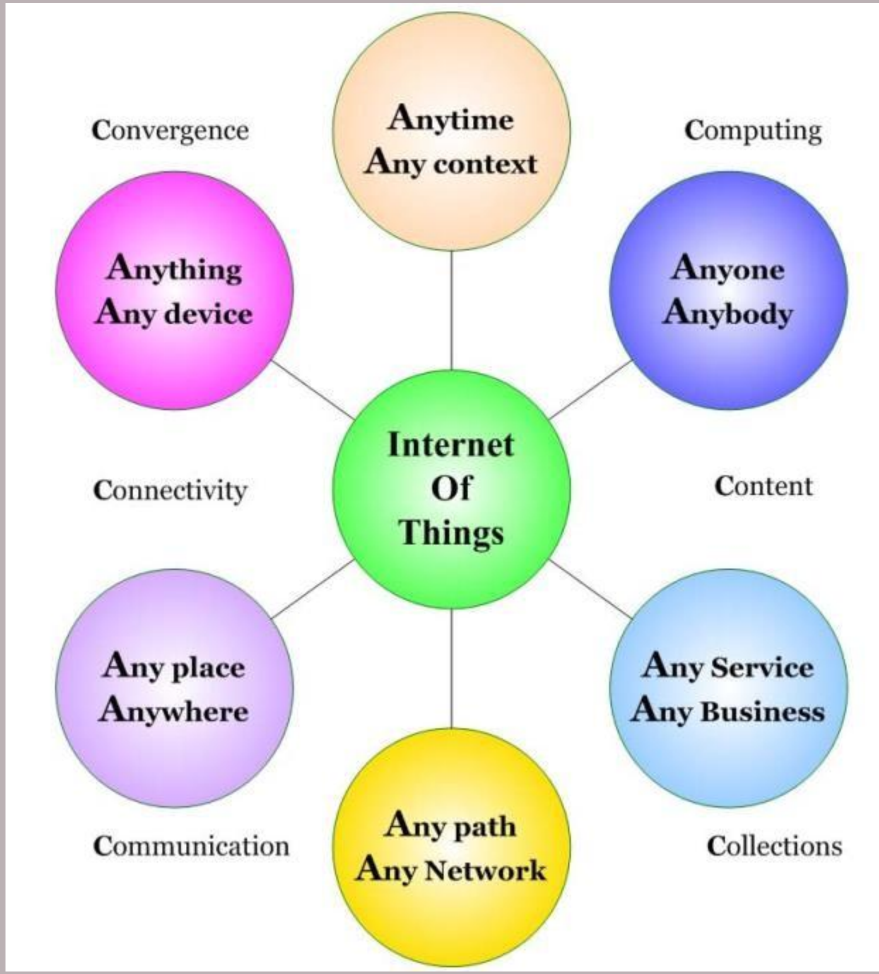
The Internet of Things allows people and things to be connected

Anytime,
Anyplace, with
Anything and
Anyone, ideally using
Any path/network and
Any service.

This implies addressing elements such as

Convergence,
Content,
Collections (Repositories),
Computing,
Communication, and
Connectivity

in the context where there is seamless interconnection between people and things and/or between things and things so the **A** and **C** elements are present and addressed.



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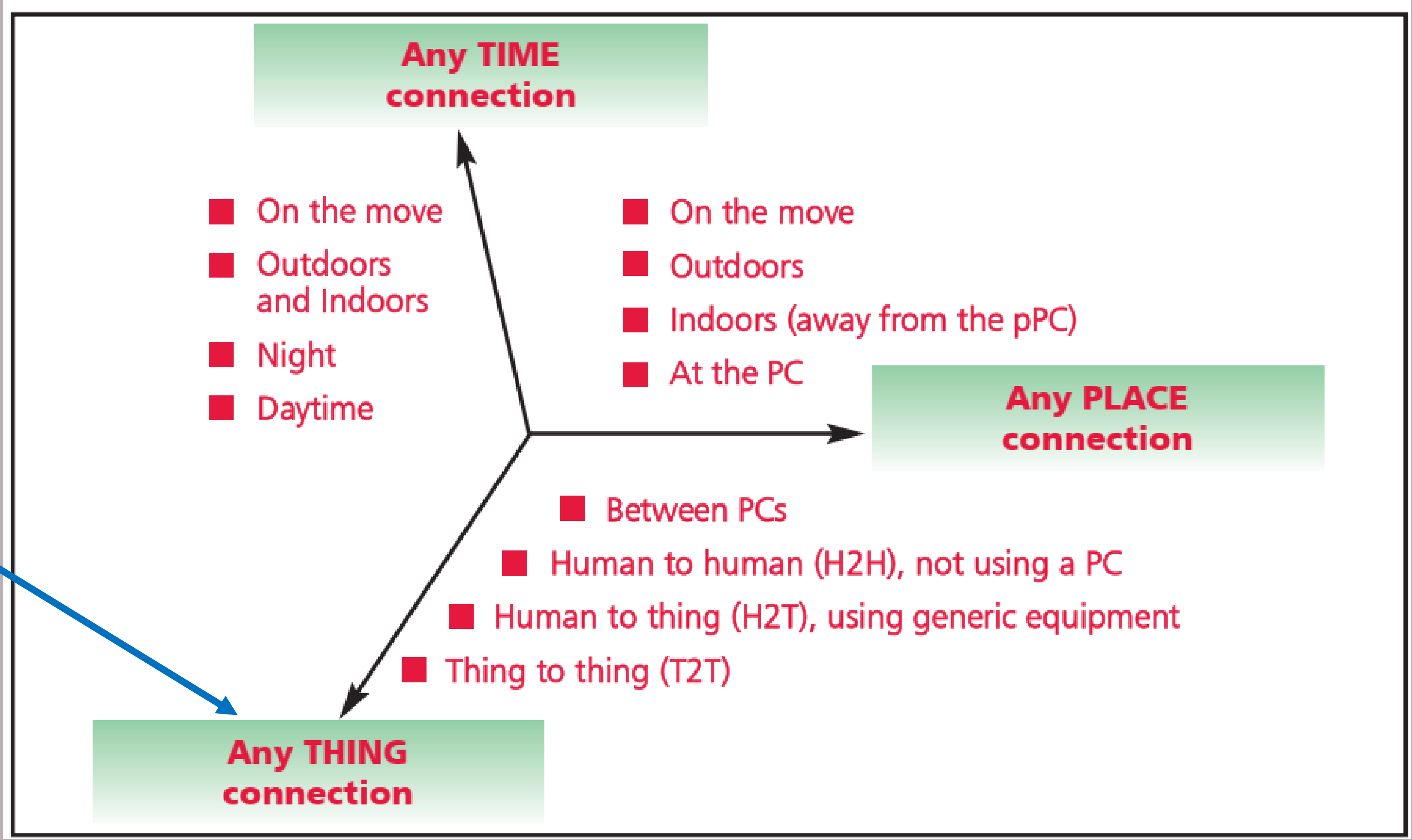
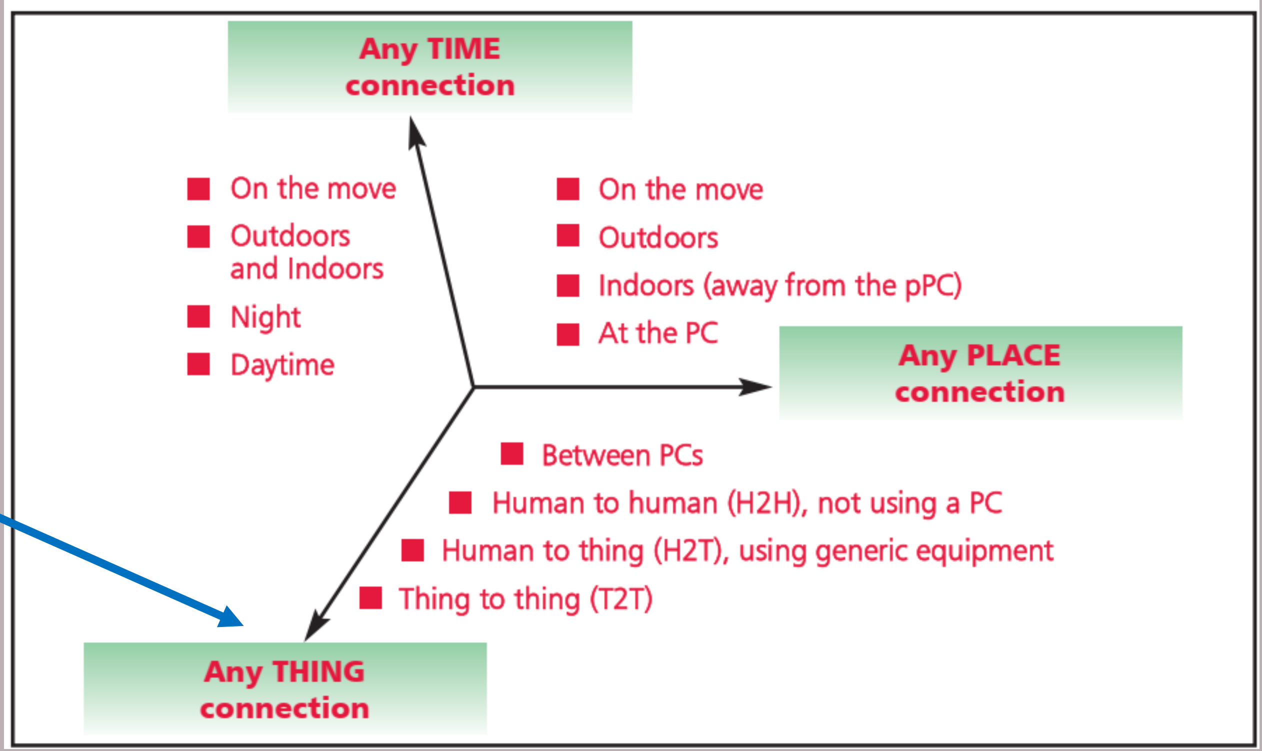
The new dimension introduced in the Internet of things: Any THING



Internet of Things

- IoT, in essence, can be seen as an addition of the third dimension named "**Thing**" to the plane of ICT world, which has been fundamentally based on two dimensions of **Place** and **Time** as shown in the figure.
- This "anything" dimension ultimately boosts the ubiquity by enabling new forms of communication of humans and things and between things themselves.



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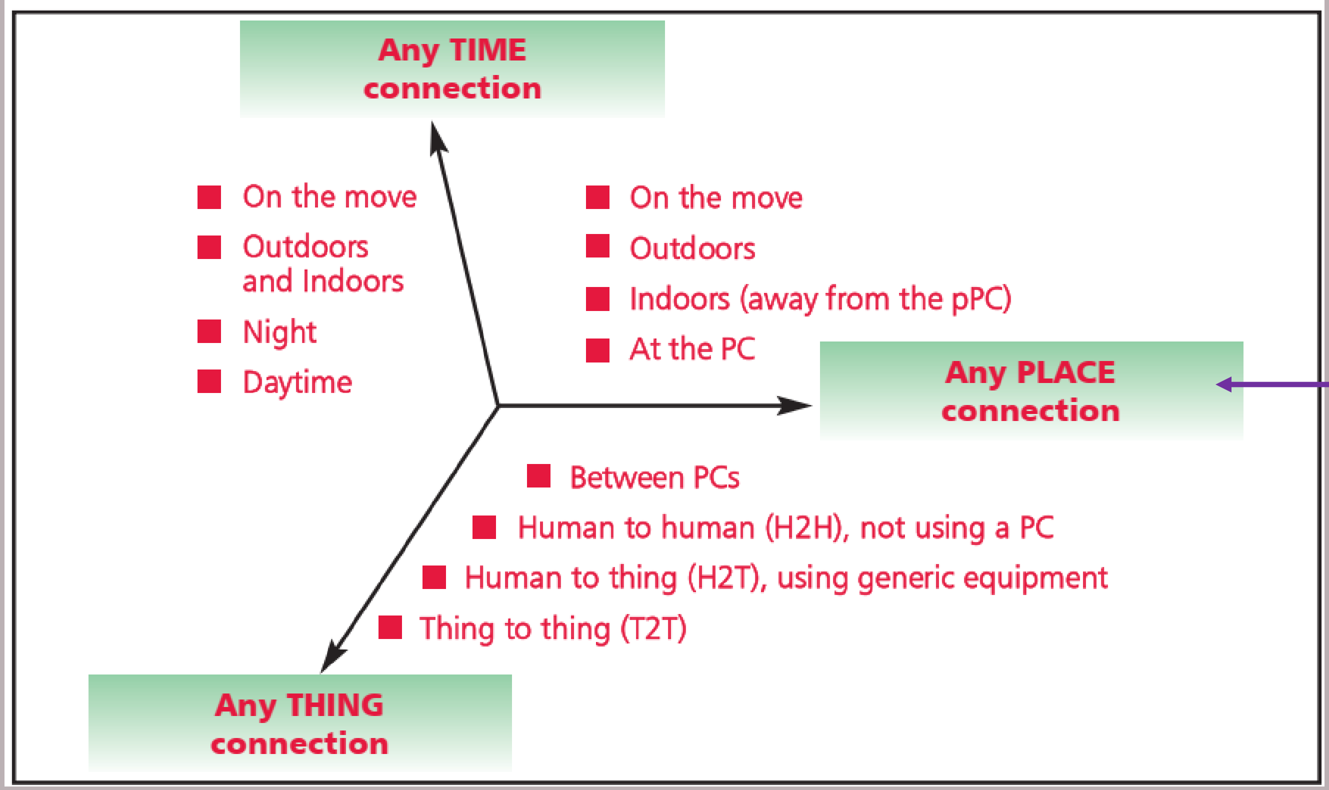


Any place connection

However, this revolution only occurred when the user sat down before a PC. Once the user got up and left the PC, this connectivity disappeared.

With the emergence of the mobile Internet service and technologies isuch as bluetooth, wireless LANs (local area networks), connectivity to a network other than by a PC came into bloom.

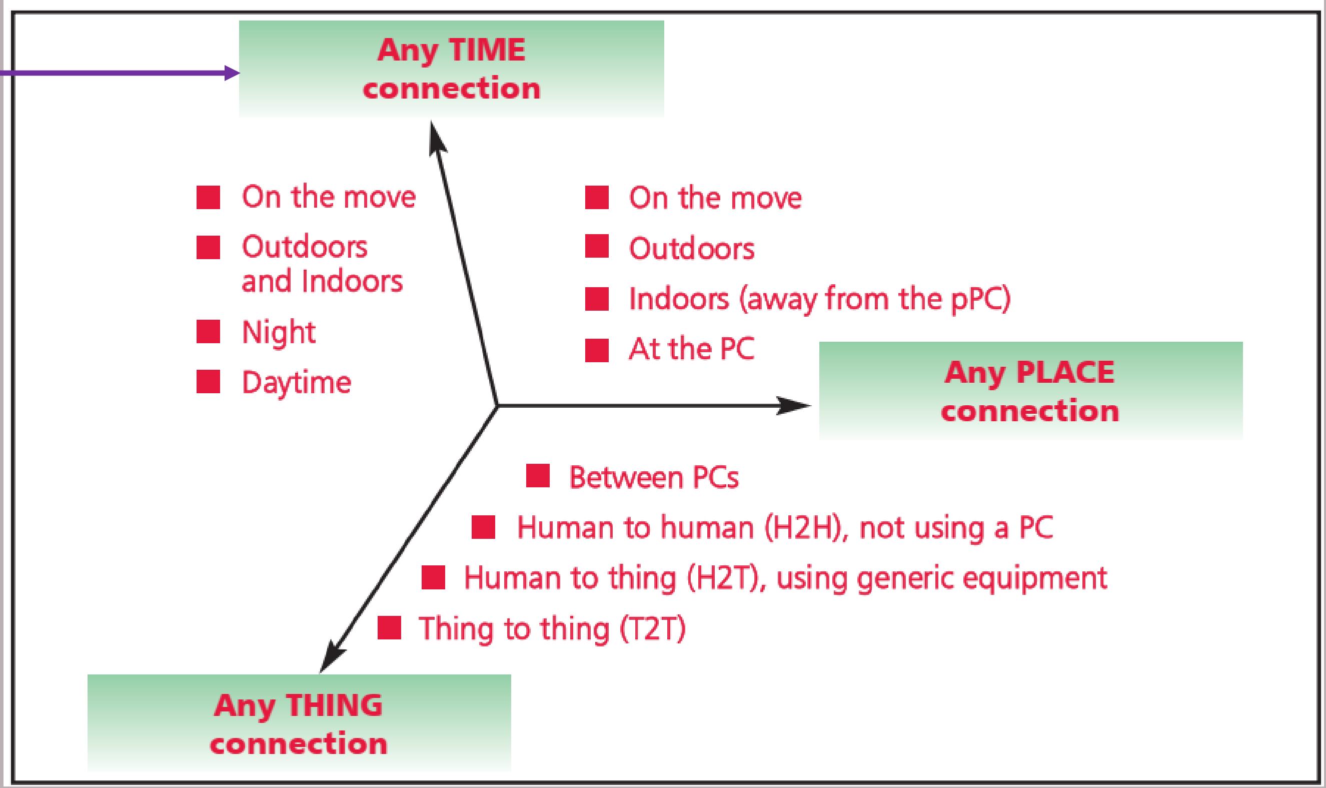
These technologies enabled connection to a network "at any place" at considerably lower cost. "Any place" examples include: (1) in front of a PC, as before; (2) in individual rooms in a building, (3) outside the home and office, and (4) in a vehicle moving at a high speed.

Internet of Things

Any time connection

The second axis relates to connecting "**at any time**."

- This means service that is always on. While these services provide an "always-on" 24 x 7, (1) access is not available when the user moves away from the PC and "always-on" connections therefore mean nothing for a user who is not in front of a PC.

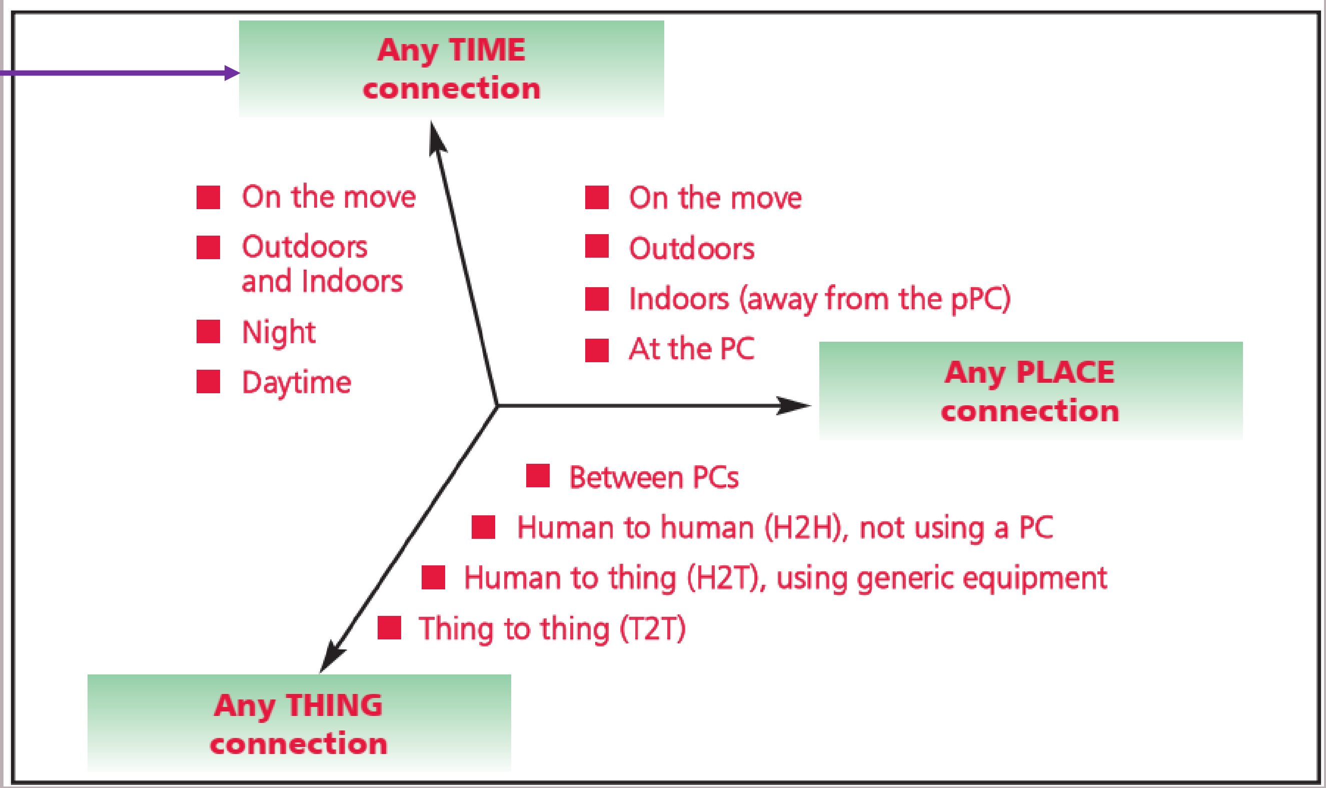


Internet of Things

Any time connection

The second axis relates to connecting "**at any time**."

- To enable connections "at any time," connections must be possible "at any place" on an always-on basis. Connections at any time can truly be realized only after an always-on status is made possible at any occasion (2) even when the user is not in front of a PC or at home or in the office, (3) even when the user is out and (4) even when the user is traveling.

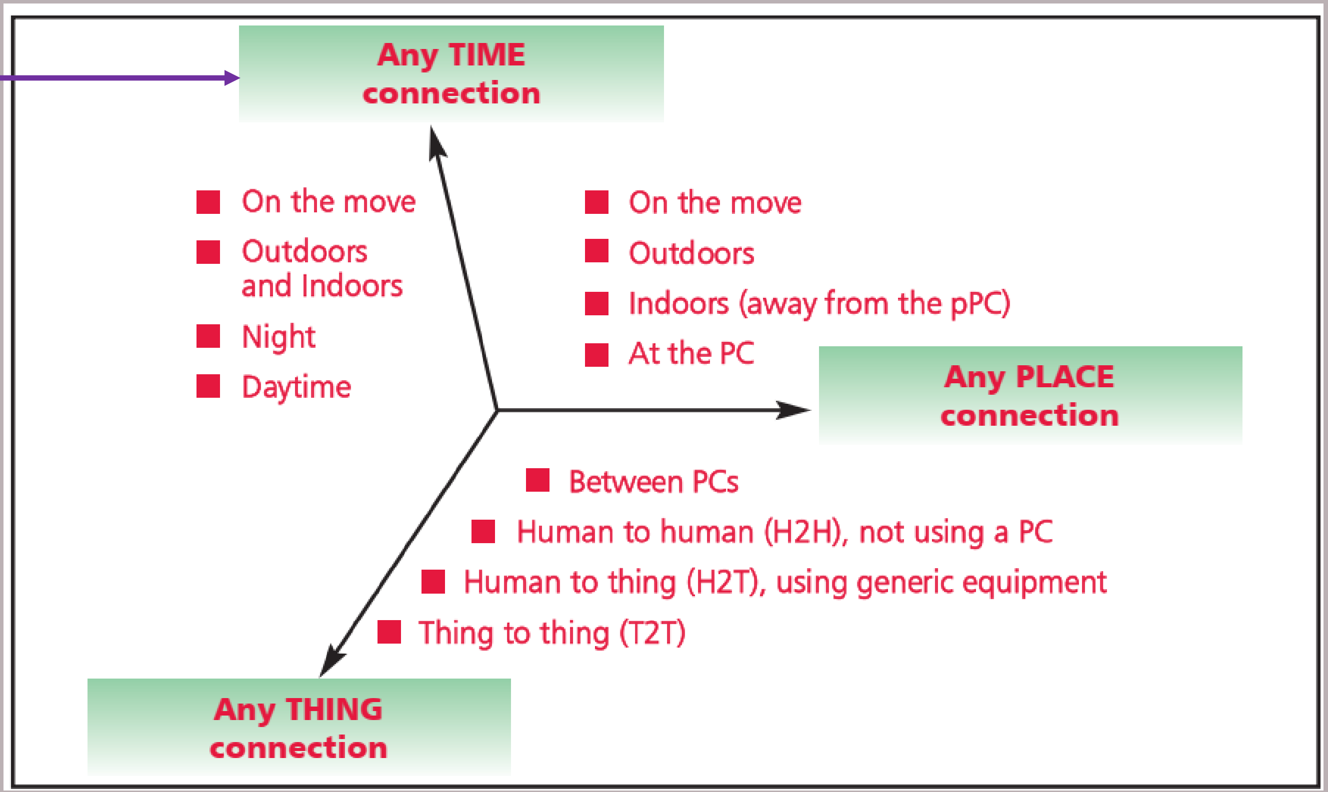


Internet of Things

Any thing connection

The third axis relates to "**any thing connection**."

The ubiquitous network enables use of a wide variety of non-PC equipment such as cell phones, game machines and car navigation equipment, in addition to desktop and mobile PCs (1). While this will facilitate "better connections" between people (2), people who have experienced the "better connection" environment between people will soon go to the next stage of requiring "better connections" between people and objects (3).



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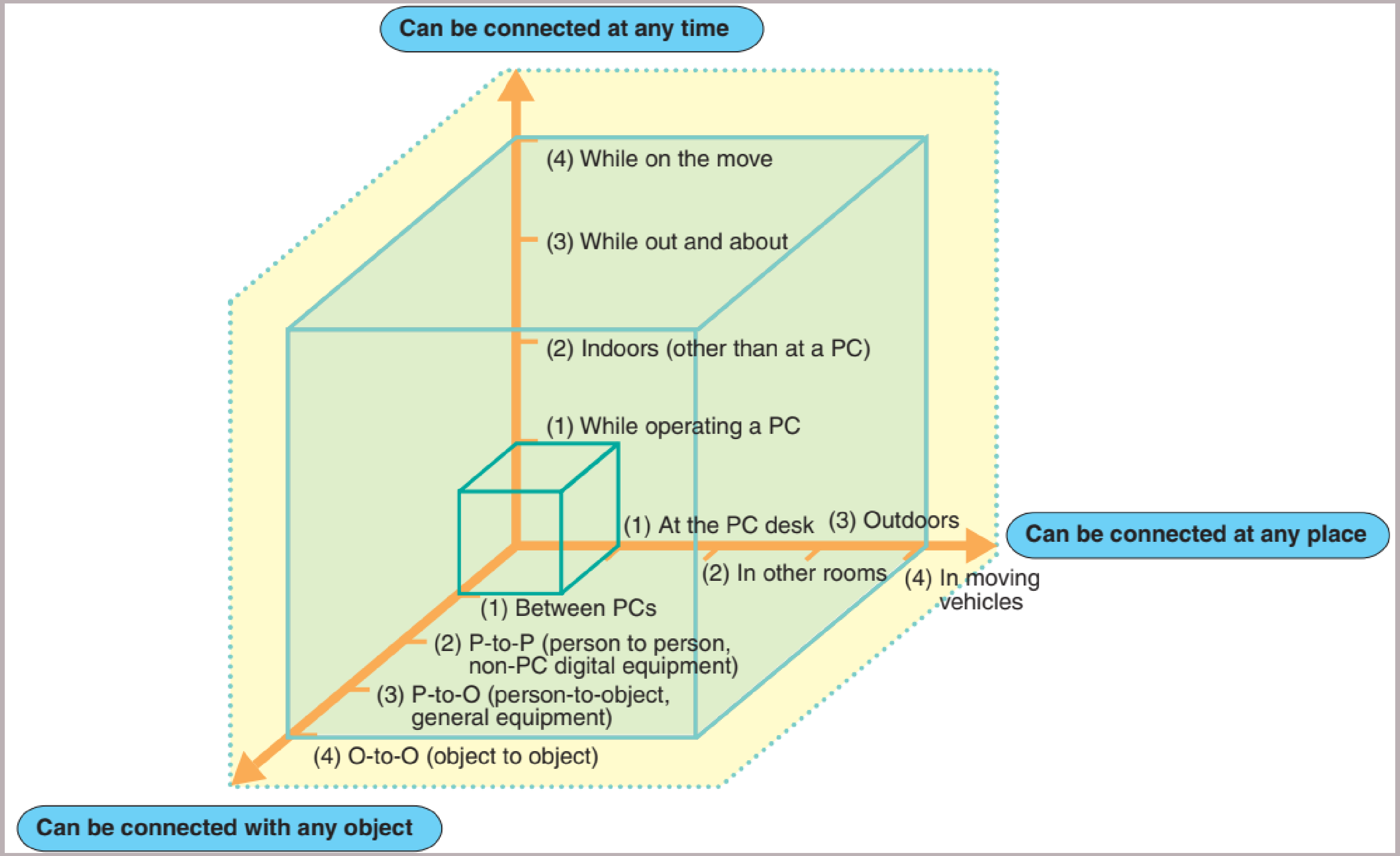


Evolutionary Stages of the Ubiquitous Network

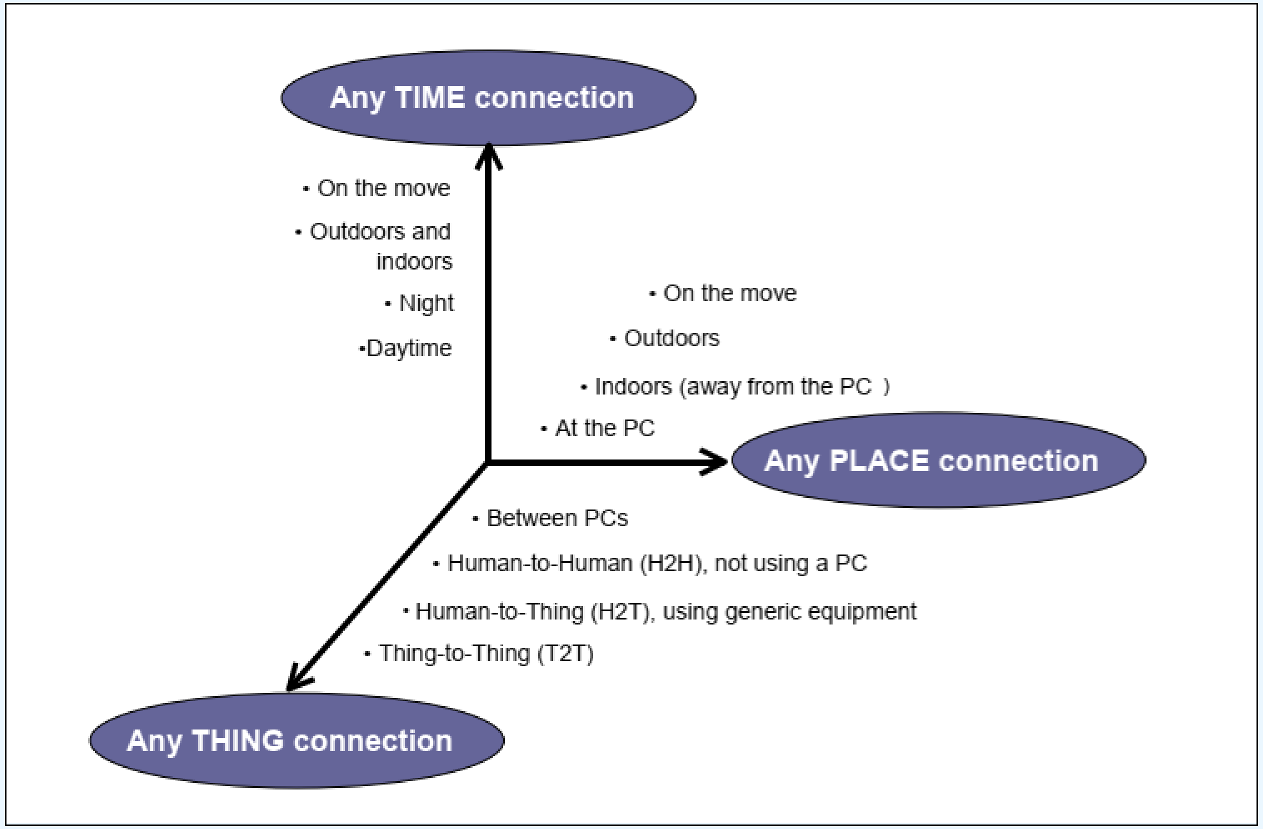
The process of ubiquitous networking can be seen as based on the three cubes shown in the figure. This process can be described as the course in which an innovation that occurred in a small cube, i.e., a communication revolution that took place in front of a PC, is further promoted and expanded to achieve the status indicated by the outermost large cube.

truly ubiquitous network: "***anytime, anywhere, by anyone and anything***".

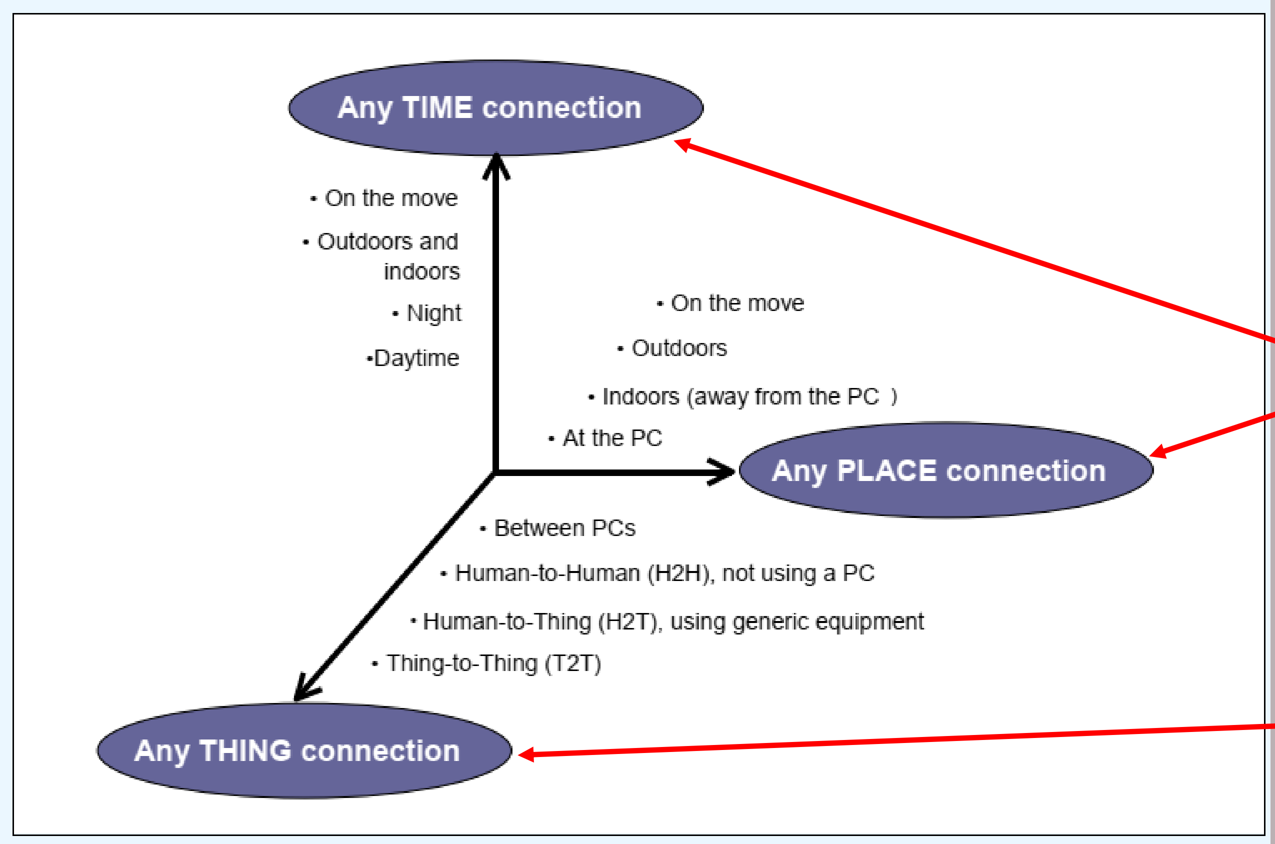
Internet of Things

Different View

- As the internet first spread, users were amazed at the possibility of contacting people and sending information across oceans and time zones, through e-mail and instant messaging, with just a few clicks of a mouse. In order to do so, however, they typically had to sit in front of a computing device (usually a PC) and dial-up to the internet over their telephone connection



Internet of Things



Different View

- Today, with mobile internet services and the deployment of higher-speed mobile networks, users can connect from almost *any location*. They can also access networks at *any time*, through *always-on* connectivity (wired and wireless broadband).
- The next step in this technological revolution is to connect inanimate objects and things to communication networks. This is the vision of a truly ubiquitous network–"*anytime, anywhere, by anyone and anything*".